DUROTECH

Waterproofing • Sealants • Adhesives

Made In Australia For Over 40 Years





After completing this course you will be able to

- Describe some of the major failures of waterproofing membranes
- Describe why current waterproofed areas leak
- Identify the potential defects hidden in waterproofing membrane and eliminate them
- Recognize the importance of using compatible full waterproofing system in building articular end use area’s
- List ways and identify what type of products can be used in particular end use area’s

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Did you KNOW?

Waterproofing is generally approximately 1% of a total construction cost.

Unfortunately it is the cause of approximately 80% of building complaints.



DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Who is DUROTECH and What we Believe



Durotech Industries is a leading manufacturer and supplier of waterproofing, sealants and adhesive technologies. Durotech Systems products have been widely used in the construction industry for the sealing and coating of Both residential, commercial and industrial buildings for over 40years.

At Durotech we don’t believe that one product can do it all but we do belive a Durotech System can.

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

What services do we provide?

- Site visits
- Local supply
- Qualified waterproofing installers
- Specification & Design Services
- Onsite and Inhouse QA and Technical Support
- Australian manufacturing for over 40 years
- Upto 20 years warranty

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

NATIONAL NATSPEC APPROVED PRODUCT PARTNER



Get access to the latest national specifications for all your local and national projects

Other affiliate partners:-

MASTER BUILDERS ASSOCIATION MEMBER
MASTER BUILDER, BETTER BUILDER

HIA you're in good hands
MEMBER

AIW Australian Institute of Waterproofing

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

The Durotech Workshop

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Points to consider when selecting a membrane system and why?

The substrate type and moisture content.

- Internal/external
- Flexibility Class
- Abrasive resistance
- Root resistance
- Curing time.
- Ease of use.
- Environmental impact
- Toxicity (safety)
- Type of clean up needed.
- One part or two part.
- Does it need reinforcing.
- Bond breaker or fillet
- Compatibility with adhesives
- Cost

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Discussion

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Things to note when waterproofing WET areas and why they are important?

1) Correct surface ***preparation*** and ***priming***

2) The membrane must not be effected by structural movement

3) ***Bond breaker*** will need to be installed

4) The use of leak proof flanges to house floor grate and taps

5) The must be allowed to cure before tiling

6) Final ***Dry film thickness*** must be between 1 and 1.5mm

7) Follow the manufacturers instructions

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Discussion

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

What is Priming and why is it so important?

Surface preparation and priming are essential processes to ensure adequate bond, even membrane film thickness and overall performance. Cleaning the substrate will remove any latience, oils, contaminants and DUST. Dust is a killer, and can result in substrate bond failure, blistering and pin holing. Primers were never meant to be a dust suppressant, and dust contamination must be removed prior to priming. Solvent based membranes require a dry, dust free substrate. Vacuuming and cleaning with a solvent based primer are recommended for dry substrates, however washing with water and applying a water based hydro epoxy will remove all dust and stabilise a damp substrate against rising damp.

Etching and non-porous primers are applied to metal angle and flashings in order to remove oils and contaminants.
Solvent primers are applied to PVC drainage control flanges and plumbing fittings in order to remove contaminants and etch the PVC.

Water based, acrylic and SBR, primers are applied to reduce the porosity in the substrate, resulting in less pinholes and a slower cure at the base of the membrane. This results in a better bond.
Most primers have an open activity window, and may need to be cleaned and reactivated once the window has lapsed.

Areas should only be primed as they are about to be coated with the membrane. The primer should dry, but not cure out to the point of losing activity or being dust contaminated.

Hardy's Scyon sheeting and other resin infused building boards require specialised primers that are compatible with the board's resin. Damp Scyon or CFC sheeting should be primed with an Epoxy primer, to stabilise rising damp and high moisture content in the sheet.

Most Epoxy primers cannot be overcoated until the day following application.

Discussion

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Liquid applied membranes fall into the following type of classifications:

Wet Film Thickness (microns)	Dry Film Thickness after Full Cure (microns)		
	100% solids	80% solids	60% solids
500	500	400	300
750	750	600	450
1000	1000	800	600
1500	1500	1200	900

Wet and dry film thickness

Knowing the solid content of liquid applied membranes allows applicators to gauge the final dry film thickness, and monitor compliance with manufacturer requirements. This is one of the more critical aspects of membrane application, as membranes that are applied too thin will compromise longation properties and reduce resistance to abrasive damage. This is one of the major concerns of all membrane manufacturers, where a large proportion of applicators continually apply liquid membranes too thin, and then blame the product for failure.

Discussion

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

What is curing of the membrane and why is it important?

Liquid applied membranes will have both a dried state and a cured state. As with primers, an activity window may apply, where a solvent may be required to reactivate the membrane for subsequent coats.

Curing of water based membranes relies on evaporation of the water out of the membrane, leaving solid resin behind as a dry film. Curing is therefore dependent on ambient humidity, temperature and airflow. High humidity and low temperatures may slow membrane dry out rates. Direct heat should never be applied to membranes while curing as it may result in blistering. Excessively hot substrates may cook the membrane, like an egg, and result in blistering and poor bond.

Acrylic membranes tend to be slow curing, as they skin on top and underneath.
Water based polyurethanes, SBR and Latex membranes are faster curing.

Discussion

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

What is Wet Dry Film Thickness and why is it so important?

Knowing the solid content of liquid applied membranes allows applicators to gauge the final dry film thickness

membranes that are applied too thin will compromise elongation properties and reduce resistance to abrasive damage

a large proportion of applicators continually apply liquid membranes too thin, and then blame the product for failure. Liquid applied membranes rely on the carrier, such as water or solvent, to evaporate leaving a final dry film. Wet film membranes, therefore, contain a percentage of solids against carrier.

Most good quality water based membranes have a solid content of 50 to 66%. A 60% solids water based membrane contains 60% polymer resin against 40% water. When 1mm (1000 microns) of wet membrane film is applied, we are left with 0.6mm (600 microns) as a dry film thickness(DFT). Water based membranes with a higher solid content are normally rapid cure, and may split or have their flexibility compromised when cured too quickly.

A good quality solvent based polyurethane membrane has a solids content of above 70%. Most are in the range of 75 to 85% solids. A PU membrane with 80% solids and 20% solvent, will cure to a DFT of 80% of wet film. When a 3mm(3000 micron) of trowel grade PU membrane is applied, it will cure to a DFT of 2.4mm (2400 microns)

How to monitor and check WFT and thickness

(insert picture of durotech WTF Slide 65)

Knowing your membrane solids content and final Dry Film Thickness (DFT) is critical in maintaining the maximum potential performance of the waterproofing system.

Measuring the membrane wet film, with a Wet Film Gauge, will give an average measurement of the wet coat and help to predict the final DFT

Discussion

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

A guide to how thickness membranes should be:

AREA	Recommended DFT	Application COATS
Lift pits	1.5 to 3	1.2 to 3mm
Basement	1.5 to 3	1.2 to 3mm
Retaining Walls	1.5 to 3	1.2 to 3mm
Podiums	1.5 to 3	1.2 to 3mm
Planter Box	1.5 to 3	1.2 to 3mm
Water feature	1.5 to 3	1.0 to 3mm
Wet area walls	2 to 3	0.6 to 1.0mm
Wet area floors	1.5 to 3	1.0 to 1.5mm
Balconies	1.5 to 3	1.2 to 2.0mm
Swimming pools	1.5 to 3	2.0 to 3mm

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

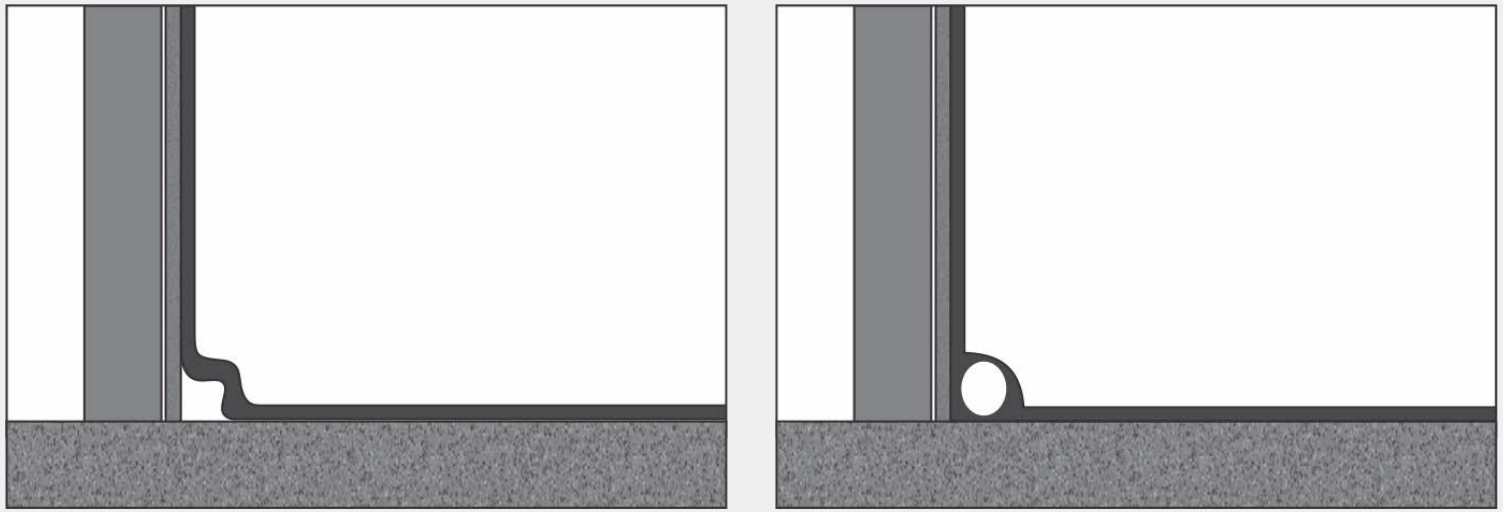
Discussion

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

How to Install Bond breakers and fillets correctly and why is it so important?

Class 11

Plastic 60% - 300% Elongation e.g. *Acrylic membrane*



Silicone with No masking tape

The liquid membrane will pool behind and underneath the closed cell foam backing rod if no tape is used.

Type II Bond Breakers are typical with cementitious membranes and bandage reinforced membranes. Reinforcement bandage can reduce a Class III membrane back to a Class II requiring a 35 mm bond breaker system. This is usually achieved by 35mm of neutral cure silicone or a sacrificial tape, where the 35mm of bandage releases from the substrate allowing independent membrane movement.

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

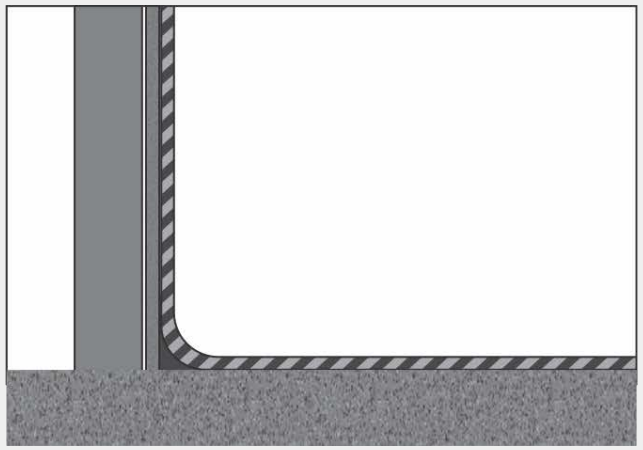
Discussion

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

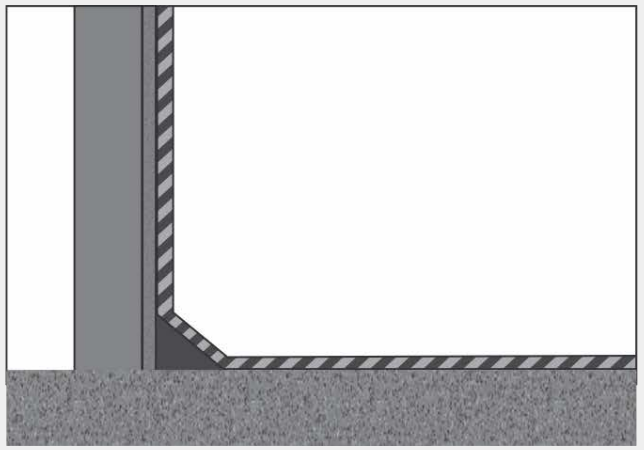
Type III Bond Breakers are functional with lining systems such as tiles or sheet cladding, where a gap of 3 to 5mm is maintained at junctions

Class III

Plastic 60% - 300% Elongation e.g. *Polyurethane*



Compatible sealant with concave finish



Compatible sealant with a flat surface

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

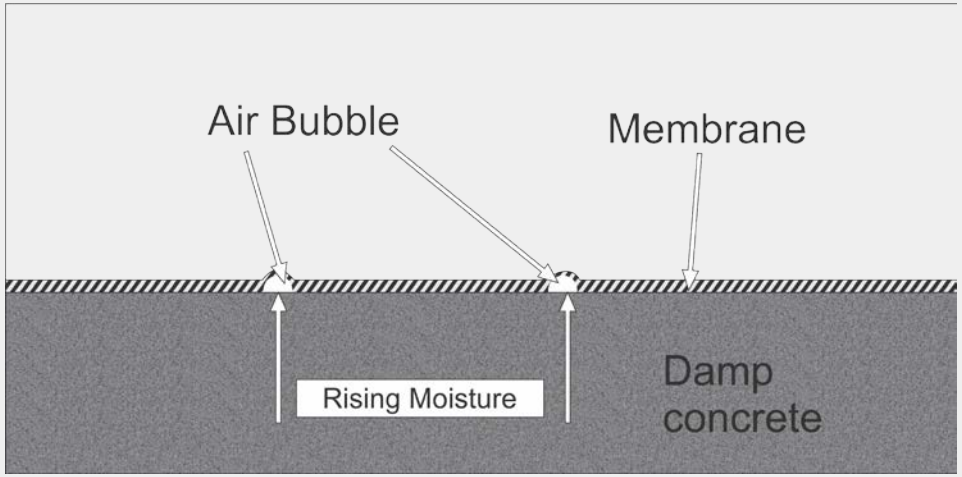
The risk of Moisture in your substrate and the problems it can cause:

The recommended content acceptable moisture should be 5% but may need to be even less depending of what type of material you are planning to use.

The solid concrete floor is most often the slowest part of the building to dry out.

Commonly the concrete is coated or topped to quickly or to soon causing the membrane to fail.

A common system used in situations where moisture is entrapped is epoxy primers. They help seal a wet or damp floor from rising moisture hence reducing bubbling of the membrane.



DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Concrete is not the hard, dense, impermeable mass we imagine it to be. It is a mineral/salt rich, porous material full of tiny channels and voids. These pathways are just right for the migration of fluids and vapours.

Nearly all coatings lose bond strength, or simply will not adhere at all, when applied to a damp, wet, or saturated surface. How dry is your concrete surface you are working on? Chances are it contains more moisture than the coating manufacturer would recommend for optimum adhesion.

High moisture content will result in rising damp and salts which will result in membrane failure through blistering and pin holing.
The application of a negative pressure membrane over the substrate will control high moisture and salts.

Negative pressure membranes may allow continuation of the membrane application where otherwise minimum curing of screeds for 7 days may need to be observed. The application of such control membranes should be incorporated in all external applications, where porous substrates are present.

Moisture metre used to assess that the membrane has dried out sufficiently. Colour change, sheen and transference of colour onto a damp rag are other indicators. Membranes and primers may be dried in a number of hours, but not fully cured for water test or functional usage for a number of days.

Discussion

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

The seven main reasons why waterproofing fails and why it happens

1. No Bond breaker
2. Uncured membrane
3. Incompatibility of bond breaker and membrane
4. Applied to thin
5. Applied to thick
6. Damage to membrane unseen pin holes
7. Inadequate surface preparation

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years



Either fill with neutral cure silicone as bond breaker, and apply membrane or, install backing rod and tape. Then bond breaker and apply membrane. Bandage can be embeded in membrane paint, left to dry and then return to apply bond breaker and membrane

Where seen membrane the sealant. It is necessary for sealant to fully cured.



DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years



Uncured Membrane with entrapped dust and dirt particles need to remove and re apply waterproofing.

Membrane is to thin can see bond breaker underneath need to apply more material. If membrane not applied to adequate thickness the membrane will not pass the manufactures performance standards.



DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years



Membrane has split in corners as its been applied to thick. Thixotropic products such as bitumen's and Cementous membranes will split when applied to thick. Must remove and re apply waterproofing system.

One pin hole can transfer through all membrane layers as trapped air, in the cavity expands and pops the drying top coat. Air must be expelled out of the pin hole and then dress the paint over. Moisture in substrate should be assessed and treated with the correct negative pressure/ rising damp primer, such as Epoxy primer or cementitious membrane.



DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Discussion

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Liquid applied membranes fall into the following type of classifications:

Membrane base	Type	Flexibility Class	Primer	Connector Sealant	Bond Breaker / Fillet	Uses & Characteristics
Water Base	Acrylic (Duromsatic ACS)	Class III	Water Based (ARW Primer)	FC Polyurethane MS Sealant (Duroseal FC) (Duroseal MS)	35mm N/C Silicone (Durosilicone)	Internal / External above ground re-emulsify, slow dry, UV stable Compatible with tiling/ render Apply above screed Apply below ***bonded*** screed ONLY
	WPU Water based polyurethane (Duromsatic P15)	Class III	Water Based (ARW Primer)	FC Polyurethane MS Sealant (Duroseal FC) (Duroseal MS)	12mm N/C Silicone (Durosilicone)	Internal / External above Ground re-emsulsify, fast dry, UV stable Compatible with tiling / render Apply above screed Apply below ***bonded*** Screed ONLY
	SBR Styrene Butadiene Rubber (Duromasti SBR Hybrid)	Class III	Water Based (ARW Primer)	MS Sealant (Duroseal FC) (Duroseal MS)	12mm N/C Silicone (Durosilicone)	Internal / External above Ground re-emulsify, fast dry, UV stable Compatible with tiling/ render Apply above screed Apply below ***bonded*** screed ONLY
	Hydro Epoxy Epoxy Primer (Duromsatic WBE Epoxy)	Class I	Hydrate into damp porous substrate			Rising damp/negative pressure limited activity window Sand surface to reactivate Inflexible / does not resist abrasion Apply above green screed or concrete Must cover with protective coating.
Cement based	Neg Pressure (Durobuild Crystalline WPM)	Class I/II	Hydrate into damp porous substrate		Cement Fillet	External Above & Below Ground Open activity Inflexible / abrasive resistant Apply above green screed or concrete Must cover with protective coating.
	2 Part (Duroflex Tank)	Class II	Water based (ARW Primer)	FC Polyurethane (MS Sealant) (Duroseal FC) (Duroseal MS)	35mm N/C Silicone (Durosilicone)	External Above & Below Ground Open activity, full immersion Potable water, resists rising damp Inflexible / abrasive resistant Apply above green screen or concrete Apply below unbonded screed

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Membrane base	Type	Flexibility Class	Primer	Connector Sealant	Bond Breaker / Fillet	Uses & Characteristics
Solvent Based	Polyurethane	Class III	Solvent primer (Hydro epoxy) (Duroprime PU) Primer (WBE Epoxy Primer)	FC Polyurethane (Duroseal FC)	FC polyurethane fillet 15 mm (Duroseal FC)	Internal / External Above & Below High flexible Ground High solids, abrasive resistant, Full immersion, very fast dry, Not Compatible with tiling/ render Apply below unbonded screed ONLY
	Polyester resin	Class I	Solvent primer Hydro epoxy	FC Polyurethane (Duroseal FC)	75mm backing rod + Vinyl tape	Internal / External Above & Below Inflexible Ground High solids, abrasive resistant Full immersion, Not Compatible with tiling / render Apply below unbonded screed ONLY
Spirits Based	Bitumenous (Duromsatic BLW)	Class II	Bitumen Paint Bitumastic Neg Pressure cement based (Durobuild Crystalline WPM)		Bitumastic fillet Cement fillet 15mm	External Below Ground Abrasive resistant Full immersion, Root repellant Not Compatible with tiling / render Apply below unbonded screed ONLY
	Coal Tar Free Polyurethane (Black Jack)	Class II	Bitumen Paint Bitumastic Neg Pressure cement based (Durobuild Crystalline WPM)		Bitumastic fillet cement 15mm	External Below Ground Abrasive resistant Full immersion, Root repellant Not Compatible with tiling / render Apply below unbonded screed ONLY.
	Bitumen Emulsion (Duromsatic BLW)	Class II	Diluted Emulsion Bitumastic (BLW) Neg Pressure Cement based (Durobuild Crystalline WPM)		Bitumastic fillet Cement fillet 15mm	External Below Ground Abrasive resistant Full immersion, Root repellant Not Compatible with tiling/ render Apply below unbonded screed ONLY

Discussion

DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years

Thanks for Listening
Any Questions?



DUROTECH
Waterproofing • Sealants • Adhesives
Made In Australia For Over 40 Years